

4 The Chemical Building Blocks of Life

Building blocks of life

- Carbon based: organic molecules
- Carbohydrates: CHO
- Lipids: CHO, water insoluble
- Proteins: CHONS, structure/function in cells
- Nucleic acids: CHONP, hereditary (genetic) information

4.1 Carbon

- Can make 4 covalent bonds¹
 - Chains
 - Straight
 - Branched
 - Ring
 - Hydrocarbons² (C, H): store energy
 - Functional groups
 - Attach to carbon
 - Alter chemical properties
 - Form macromolecules
 - Sapoteton

4.2 Carbohydrates

- Principally CHO (rare N, S and P)
 - 1C:2H:1O ratio
 - Energy rich (many C-H bonds)
- Monosaccharides (principal: glucose³)
 - Simple sugars
 - Principle formula: C₆H₁₂O₆
 - Form rings in water solution
- Disaccharides (sucrose, lactose)
- Polysaccharides (starch, glycogen, cellulose, chitin)

¹ <http://en.wikipedia.org/wiki/covalent%20bonds>

² <http://en.wikipedia.org/wiki/Hydrocarbons>

³ <http://en.wikipedia.org/wiki/glucose>

4.3 Stereoisomers

- Bond angles of carbon point to corners of a tetrahedron
- When 4 different groups are attached to a carbon, it is asymmetric, leading to various types of isomerism
 - Stereoisomers: (D, L)
- Same chemical properties
- Different biological properties
- D sugars, L amino acids

4.4 Lipids

- C-H bonds (nonpolar) instead of C-OH bonds as in carbohydrates
 - High energy
 - Hydrophobic (insoluble in water)
- Categories
 - Fats: glycerol and three fatty acids
 - Phospholipids: primary component of membranes
 - Prostaglandins: chemical messengers (hormones)
 - Steroids: membrane component; hormones
 - Terpenes: pigments; structure

4.4.1 Fatty acids

- Hydrocarbon chain
 - Even number of C, 14->20
 - Terminates in carboxyl group
- Saturated: contain maximum number of hydrogens (all single bonds); maximum energy
- Unsaturated: one or more double bonds
 - Usually higher melting point
 - Many common oils are polyunsaturated

4.5 Proteins

- Polymer of amino acids
 - 21 different amino acids found in proteins
 - Sequence of amino acids determined by gene
- Amino acid sequence determines shape of molecule
 - Linked by peptide bond (covalent)
- Functions
 - regulate chemical reactions and cell processes [enzymes]
 - form bone and muscle; various other tissues
 - facilitate transport across cell membrane [carrier proteins]

- fight disease [antibodies]
- Motifs: folding patterns of secondary structure
- Domains: structural, functional part of protein often independent of another part; often encoded by different exons
- Shape determines protein's function

4.5.1 Amino acids

- 21 commonly found in proteins
 - 21st is selenocysteine, not mentioned in text
- Common structure
 - Amino group: NH_2
 - Carboxyl group: COOH
 - R group- 4 different kinds of R groups
 - acidic
 - basic
 - hydrophilic (polar)
 - hydrophobic (nonpolar)
- Confer individual properties on amino acids
- List of amino acids⁴

4.5.2 Structure

- Primary structure: the amino acid sequence
 - Determines higher orders of structure
 - Critical for structure and function of protein
- Secondary: stabilized by intramolecular hydrogen bonding
 - helix
 - sheet
- Tertiary: folding, stabilized by ionic bonds (between R groups), hydrogen bonding, van der Waal's forces, hydrophobic interactions
- Quaternary: ≥ 2 polypeptides

4.5.3 Function

- Requires proper folding, cofactors, pH, temperature, etc.
- Proteins are often modified after synthesis
 - Chemical modification
 - Addition of heme groups (hemoglobin, cytochrome)
- Denatured proteins can not function properly
- Proteins are degraded by proteasome as part of constant turnover of cell components

4 <http://en.wikipedia.org/wiki/Amino%20acid%23List%20of%20amino%20acids>

4.6 Hereditary (Genetic) information

- Nucleic acids
 - DNA: deoxyribonucleic acid
- Hereditary information of all cells
- Hereditary information for many viruses
 - RNA: ribonucleic acid
- Hereditary information of certain viruses (HIV⁵)
- Intermediate in gene expression
- Composed of nucleotides
 - Ribonucleotides
 - Deoxyribonucleotides

4.6.1 RNA DNA origin

- Which came first?
- Paradox: DNA encodes protein necessary for its own replication
- Discovery of catalytic RNA by Cech and Altman suggested that RNA might have been first self-replicating molecule
- DNA evolved as more stable type of storage molecule

This text is based on notes very generously donated by Dr. Paul Doerder, Ph.D., of the Cleveland State University.

Proteins: Their building block is amino acids. The bond connecting 2 of the amino acids together are called peptide bonds. One of these bonds makes a monopeptide, two a dipeptide, and any more than that makes a polypeptide.

5 <http://en.wikipedia.org/wiki/HIV>